# An obstruction to multiplying six-contact functions

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**Abstract**

The six rational contacts used in the weak-*abc* construction admit a rational involution which changes the sign of their product modulo squares. We give positive rational scales of unbounded height for which the product of the two contact functions has even valuation at every finite prime. Consequently the odd radical of that product cannot by itself control the scale height on the unrestricted rational domain. This obstruction is independent of any *abc* conjecture. It does not assert the existence of witnesses for the remaining source equations or for their prescribed eighth-power scale.

## 1 The contact function and its involution

For $u \in \mathbb{Q}^\times$, define its odd radical by

$$\mathrm{Odd}(u) = \prod_{\substack{p \text{ prime} \\ v_p(u) \text{ odd}}} p.$$

Here negative valuations are included. Thus $\mathrm{Odd}(u) = 1$ if and only if $u$ is a rational square or the negative of a rational square. For $u = a/b$ in lowest terms with $b > 0$, put $h(u) = \log \max(|a|, b)$.

Let

$$\Theta = \{0, -1/2, -1/4, -2/5, -1, -1/3\}.$$

Put

$$F(t) = \prod_{\theta \in \Theta} (t - \theta) = \frac{t(t+1)(2t+1)(3t+1)(4t+1)(5t+2)}{120},$$

and define

$$R(t) = \frac{F(t)}{(t-1)^8}, \qquad \phi(t) = -\frac{2t+1}{3t+2}.$$

These are the six contacts and the contact function of the weak-*abc* construction in [1]. The rational map $\phi$ is an involution of the projective line. It exchanges the contact pairs

$$\{0, -1/2\}, \qquad \{-1/4, -2/5\},$$

and fixes $-1$ and $-1/3$.

**Lemma 1.1.** *For every rational $t$ for which the displayed expressions are defined,*

$$F(\phi(t)) = -\frac{F(t)}{(3t+2)^6}, \qquad R(t)R(\phi(t)) = -\left(\frac{F(t)(3t+2)}{(t-1)^4(5t+3)^4}\right)^2. \tag{1}$$

*In particular the second product has even valuation at every finite prime whenever it is nonzero.*

*Proof.* Write $N = -2t - 1$ and $D = 3t + 2$. Then

$$N + D = t + 1, \qquad 2N + D = -t, \qquad 3N + D = -3t - 1,$$
$$4N + D = -5t - 2, \quad 5N + 2D = -4t - 1.$$

Consequently

$$N(N + D)(2N + D)(3N + D)(4N + D)(5N + 2D) = -120F(t).$$

Dividing by $120D^6$ proves the first identity. Also $\phi(t) - 1 = -(5t + 3)/(3t + 2)$, whence

$$R(\phi(t)) = -\frac{F(t)(3t + 2)^2}{(5t + 3)^8}.$$

Multiplying by $R(t)$ proves the second identity. Every finite valuation of the negative of a nonzero square is even. □

## 2 Unbounded positive scales

**Theorem 2.1.** *For every integer $m \geq 5$, put*

$$\lambda_m = -\frac{1}{m}, \qquad a_m = \frac{m(m - 2)}{2m - 3}.$$

*Then $a_m > 1$, both contact functions below are defined and nonzero, and*

$$a_m\lambda_m = \phi(\lambda_m), \qquad \mathrm{Odd}\big(R(\lambda_m)R(a_m\lambda_m)\big) = 1.$$

*Moreover $h(a_m) \longrightarrow \infty$ as $m \longrightarrow \infty$. Hence no constants $C, C_0$ give an inequality*

$$h(a) \leq C \log \mathrm{Odd}\big(R(\lambda)R(a\lambda)\big) + C_0 \tag{2}$$

*for all positive rational $a > 1$ and all rational $\lambda$ for which the two contact functions are defined and nonzero.*

*Proof.* Direct substitution gives

$$\phi(-1/m) = -\frac{m - 2}{2m - 3} = a_m\lambda_m, \qquad a_m - 1 = \frac{(m - 1)(m - 3)}{2m - 3} > 0.$$

For $m \geq 5$,

$$-\frac{1}{4} < \lambda_m < 0, \qquad -\frac{1}{2} < \phi(\lambda_m) < -\frac{2}{5}.$$

Neither open interval contains a contact or the pole 1 of $R$. Thus the product is nonzero and defined. Lemma 1.1 makes its odd radical equal to one. Finally $a_m > (m - 2)/2$, so $h(a_m) \geq \log a_m \longrightarrow \infty$. The right-hand side of (2) would be the fixed constant $C_0$ on this family, a contradiction. □

Removing any fixed finite set of primes from the definition of Odd leaves this obstruction unchanged. Multiplication or division of the product by a nonzero rational square also leaves it unchanged.

## 3 Scope and exact verification

The theorem concerns an unrestricted contact-height inequality. In the weak-*abc* defining source, the scale is specifically $a = \beta^8 + T(n)^8$, where $T(n)$ is the elliptic trace, and the contact expressions are accompanied by other equations and guards. The family above is not asserted to satisfy that scale condition or those equations. Thus it does not give a counterexample to the thirteen-variable theorem, or exclude every possible construction with one pole predicate. It shows that a direct product argument needs additional arithmetic information about the scale or the common contact primes. The obstruction holds whether or not *abc* is assumed.

The standard-library program

`computation/pole_parameters/check_six_contact_product.py`

checks the homogenized degree-six identity, the involution matrix, the contact permutation, and the parameterized scale identities by exact polynomial arithmetic. It also evaluates the resulting negative-square identity for every integer $5 \le m \le 200$ with exact fractions. The polynomial identities and the proof above establish the infinite family; the finite evaluations are additional checks, not the basis for that assertion.

## References

[1] *Thirteen Variables from a Weak abc Inequality*, companion paper, 9 November 2026, `conditional13_20261109_v16.tex`.